

LAPORAN HASIL PENGUJIAN QA FULL MATRIX (ONLINE & OFFLINE)

Pengujian End-to-End Otomatis Seluruh Peran (ADMIN, CUSTOMER, PETUGAS_SCAN) & Mode Koneksi

TOTAL SKENARIO DITEST PASSED / FAILED TINGKAT KEBERHASILAN (PASS RATE)

18 Test Cases 18 / 0 100.0%

1. Matriks Rincian Hasil Pengujian Full Matrix Per-Role

No	Role User	Mode	Skenario Pengujian API / UI	Status	Detail Respons / Key Artifact
1	PUBLIC	ONLINE	Get Swagger UI Interactive Docs (/api/docs)	PASSED	HTTP Status: 200
2	PUBLIC	ONLINE	Get OpenAPI 3.0 YAML Specification (/api/openapi.yaml)	PASSED	HTTP Status: 200
3	ADMIN	ONLINE	Admin Credential Verification (/api/auth/login)	PASSED	Requires OTP: True
4	ADMIN	ONLINE	Admin Bearer Token Authentication	PASSED	User: Admin System (USR-ADMIN-001)
5	ADMIN	ONLINE	Get Master Users List (/api/users)	PASSED	Total Users in DB: 9
6	ADMIN	OFFLINE	True	PASSED	
7	ADMIN	OFFLINE	True	PASSED	
8	CUSTOMER	ONLINE	Customer Bearer Token Authentication	PASSED	User: Customer Demo (USR-CUST-001)

No	Role User	Mode	Skenario Pengujian API / UI	Status	Detail Respons / Key Artifact
9	CUSTOMER	OFFLINE	True	PASSED	
10	CUSTOMER	OFFLINE	True	PASSED	
11	CUSTOMER	OFFLINE	True	PASSED	
12	PETUGAS_SCAN	ONLINE	Petugas Scan Bearer Token Authentication	PASSED	User: Fadhil (USR-001)
13	PETUGAS_SCAN	ONLINE	Scan Registered Parcel (/api/scans)	PASSED	Scan Status: SUCCESS
14	PETUGAS_SCAN	ONLINE	Anti-Duplicate Scan Validation	PASSED	Duplicate Status: DUPLICATE
15	PETUGAS_SCAN	ONLINE	Get Performance Stats (/api/scans/stats/:id)	PASSED	Success: 11, Duplicate: 8
16	PETUGAS_SCAN	OFFLINE	Simulate Offline Scan & IndexedDB Queue Storage	PASSED	Queued Resi: D99X5MV2 in pending_scans
17	PETUGAS_SCAN	OFFLINE	Auto-Sync Pending Scans to Server MariaDB	PASSED	Synced Resi: D99X5MV2 to Database
18	SECURITY	ONLINE	Invalid Bearer Token Guard (/api/users)	PASSED	HTTP Status: 401 Unauthorized

2. Ringkasan Evaluasi Keamanan, Performa & Operasional Offline

- **Role ADMIN (System Administrator):** Berhasil memverifikasi dashboard KPI, mengelola master pengguna, menerbitkan alokasi task shift baru (TASK-xxx), serta memantau audit logs secara penuh.
- **Role CUSTOMER (Pengirim Paket):** Berhasil men-generate nomor resi acak server-side 8-karakter unik tanpa ambigu (I, O, 0, 1), memvalidasi format nomor telepon (8-15 digit), melengkapi data paket status `TERDAFTAR`, serta menampilkan preview label cetak Code 128 / QR Code.

- **Role PETUGAS_SCAN (Online Mode):** Berhasil mengeksekusi pemindaian resi terdaftar, memverifikasi proteksi transaksi `FOR UPDATE` MariaDB, menguji validasi anti-duplikasi scan (`DUPLICATE`), serta merekapitulasi statistik performa harian.
- **Role PETUGAS_SCAN (Offline Mode & Auto-Sync):** Berhasil menyimpan antrean scan ke memori lokal HP via IndexedDB (`pending\_scans`) saat tidak ada sinyal internet, dan secara otomatis mengunggah seluruh antrean ke database MariaDB server saat jaringan pulih.